ALICE L WALTON SCHOOL OF MEDICINE

PDE–5 Inhibitors

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Describe the mechanism of action of PDE-5 inhibitors in the context of treating erectile dysfunction.
2. Compare pharmacokinetic properties of PDE-5 inhibitors and their clinical implications.
3. Explain the adverse effects associated with the use of PDE-5 inhibitors.
4. Discuss the [illegible] associated with nitrates.

CASE CONNECTION

Let's bring back a familiar friend from the nitrate brick, JD.

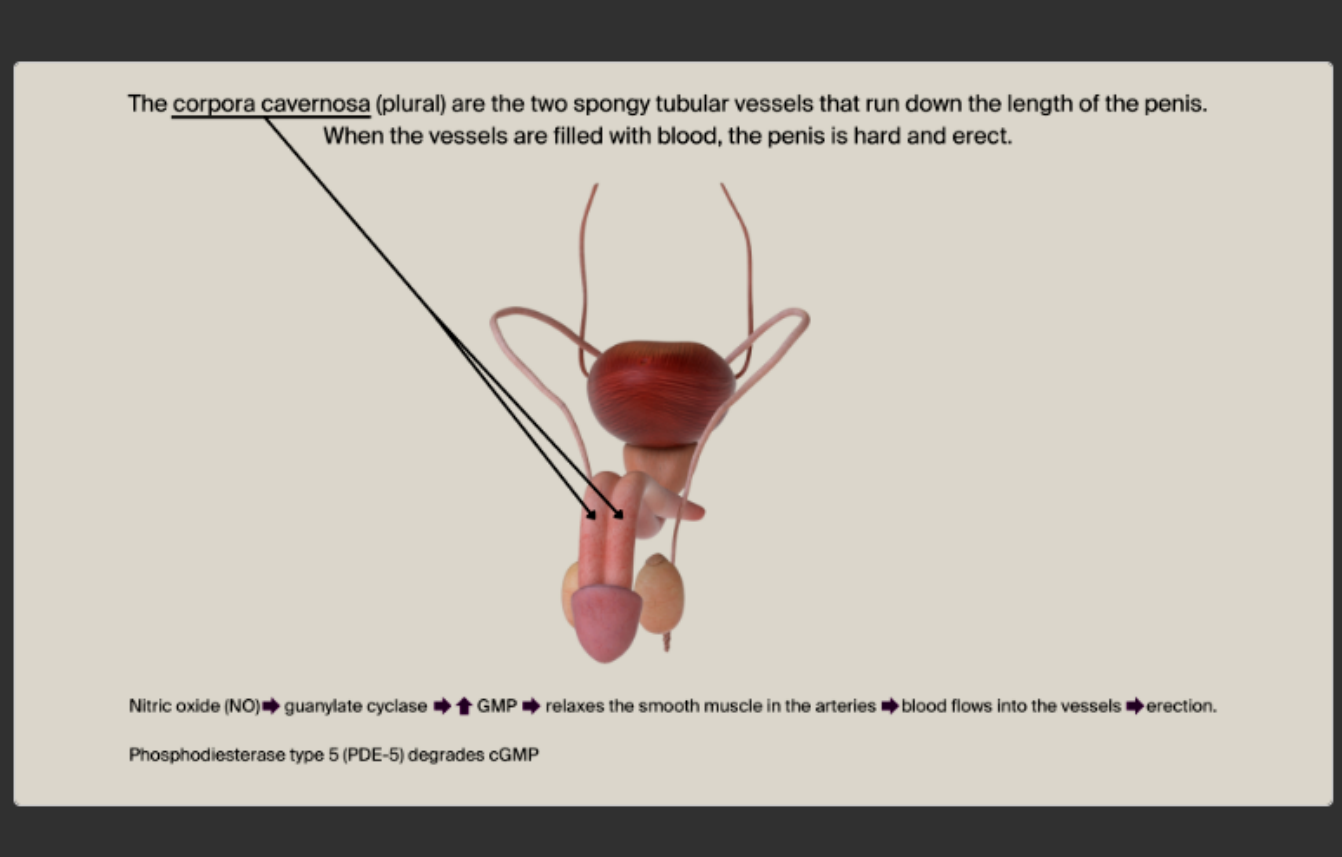
Remember, JD is a 65-year-old male who had unstable angina and was admitted to your medicine team. He has a history of hypertension, hyperlipidemia, chronic stable angina, and a myocardial infarction 5 years ago. He takes many different medications for these indications, most notably isosorbide mononitrate 60 mg daily.

You are following up with JD in the clinic a month after he is discharged from the hospital. He now expresses concerns about erectile dysfunction (ED) over the past year. He describes difficulty achieving and maintaining erections sufficient for satisfactory intercourse. This issue has caused distress and negatively impacted his relationship with his spouse.

Are there any contraindications to medication therapy to treat JD's ED? What adverse effect could occur if he receives a PDE-5 inhibitor?

GO TO CONCLUSION ↓

The male sexual response is a very complex process that involves muscles, nerves, blood vessels, hormones, and the brain. A problem with any of these can lead to erectile dysfunction (ED), which is the inability to achieve or maintain an erection for sexual intercourse or pleasurable sexual activity. The prevalence of ED incr[illegible]ows the mechanism of erection under normal circumstances. Recall, from the nitrate brick, the same process of nitric oxide (NO) activating guanylyl cyclase, which converts GTP to cGMP. The net result is relaxation of smooth muscles and increased blood flow.


The corpora cavernosa (plural) are the two spongy tubular vessels that run down the length of the penis. When the vessels are filled with blood, the penis is hard and erect.
Nitric oxide (NO) ➡ guanylate cyclase ➡ ⬆ GMP ➡ relaxes the smooth muscle in the arteries ➡ blood flows into the vessels ➡ erection.
Phosphodiesterase type 5 (PDE-5) degrades cGMP

The most common treatment for ED without a clear correctable cause is phosphodiesterase type 5 (PDE-5) inhibitors. For the remainder of the brick we will focus on this medication class.

CLINICAL CORRELATION

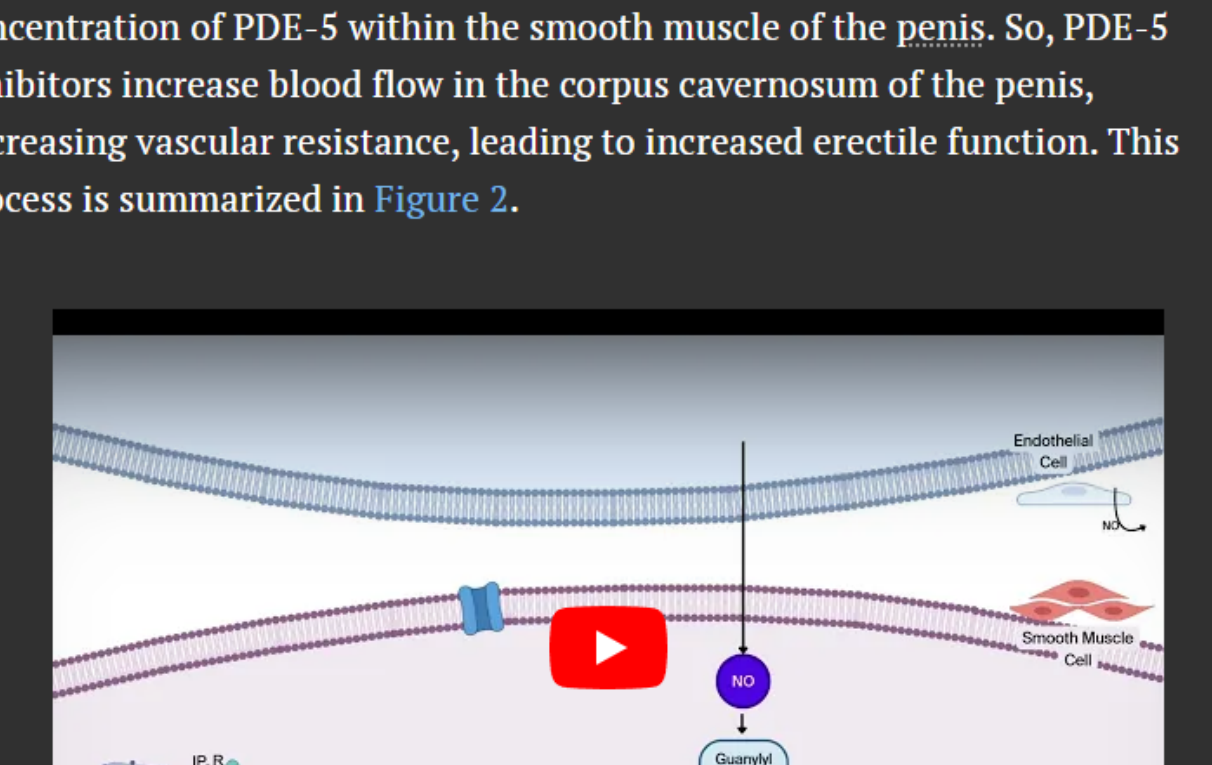
PDE-5 inhibitors are [illegible]on, however, they are also a mainstay of treatment for pulmonary arterial hypertension (PAH). In fact, sildenafil was first studied only for the indication of PAH, but when researchers found male patients were experiencing erections as a 'side effect'. Research then began focusing on this 'side effect' and sildenafil also received the indication for erectile dysfunction.

What enzyme does nitric oxide (NO) activate to facilitate erection?

How do PDE–5 inhibitors work? (LO#1)

Recall from the nitrate brick, as nitrates are converted to nitric oxide this increases the amount of circulating cGMP. Then cGMP modulates the activity of phosphodiesterases (Primarily 2, 3, and 5) in smooth muscles which increases vasodilation, i.e., blood flow.

The medication class PDE–5 inhibitors work on the same pathway, but a different area. Under normal circumstances, PDE-5 degrades cGMP, which would lead to less smooth muscle relaxation and decrease blood flow. PDE-5 inhibitors block this enzyme and reduce cGMP breakdown. This leads to prolonged smooth muscle dilation in response to nitric oxide. There is a large concentration of PDE-5 within the smooth muscle of the penis. So, PDE-5 inhibitors increase blood flow in the corpus cavernosum of the penis, decreasing vascular resistance, leading to increased erectile function. This process is summarized in Figure 2.



What is the primary action of PDE-5 inhibitors in treating ED?

What is the difference between PDE–5 inhibitor medications? (LO#2)

There are four PDE-5 inhibitors available to treat ED in the United States, sildenafil, tadalafil, vardenafil, and avanafil. Sildenafil and tadalafil are by far most commonly used. Each differ in their pharmacokinetic profiles, making them better suited for some patients/ situations. A comparison of the four medications is summarized in Table 2, however, there are two considerations that are discussed in further detail.

Table

Characteristic	Sildenafil	Tadalafil	Vardenafil	Avanafil
Onset of action	1 hour	0.5 hour	1 hour	0.5 hour
Duration of action	4 hours	36 hours	7 hours	6 hours
Absorption affected by fatty meals	Yes	No	Yes	Yes
Renal Adjustmen[illegible]	[illegible]	[illegible]	Yes	Yes

Tadalafil has the longest time frame of efficacy, or duration of action, of up to 36 hours. This gives the patient a much larger window for sexual activity, relative to the other PDE-5 inhibitors. This is owed to the extended half-life of tadalafil, but, as we will discuss below in the adverse effects, it can present a challenge in a patient with cardiovascular comorbidities.

All of the PDE-5 inhibitors are eliminated renally. This means that in a patient with decreased renal function the doses must be lowered. Tadalafil, again because of its long half-life, is the most concerning for toxicity due to accumulation in patients with altered kidney function.

Why must tadalafil doses be lowered in patients with decreased renal function?

What are the adverse effects of PDE–5 inhibitors? (LO#3)

Similar to the nitrate medications. The most common adverse effects associated with PDE-5 inhibitors are headache, flushing, and dizziness, again, due to the vasodilation that occurs. PDE-5 inhibitors can also cause hypotension. This is generally mild and not noticeable, however, when combined with other medications (see section below) it can be severe.

Dyspepsia (i.e., indigestion or upset stomach) may also occur, this is due to relaxation of the smooth muscles of the lower esophageal sphincter. The relaxation of these muscles allows gastric acid to reflux into the esophagus causing dyspepsia.

Patients taking PDE-5 inhibitors should also be aware of priapism, which is an erection lasting over four hours. If this occurs, the patient should seek medical attention, as it often requires a procedure to drain blood from the corpus cavernosum.

What are the common adverse effects of PDE-5 inhibitors?

What are the drug interactions and contraindications for PDE–5 inhibitors? (LO#4)

PDE-5 inhibitors have some important interactions and contraindications to be aware of. Two PDE-5 inhibitors should never be used at the same time, due to the risk of profound hypotension. Also, the most concerning interaction is between nitrates and PDE-5 inhibitors. These two classes should not be taken at the same time.

PDE-5 inhibitors and nitrates act synergistically to cause profound increases in cGMP, significant vasodilation, and, ultimately, dramatic reductions in blood pressure. PDE5 inhibitors should therefore not be prescribed to patients receiving any form of nitrate; in prescribing nitrates, the patient should be warned that PDE5 inhibitors and nitrates must not be used concurrently and that no PDE-5 inhibitor should be used in the 24 h prior to initiating nitrate therapy. A period of longer than 24 h may be needed following administration of tadalafil, due to its prolonged t1/2.

Caution must also be exercised when giving PDE-5 inhibitors with other blood pressure lowering agents (including alpha-1 blockers), as profound hypotension could occur.

Sildenafil, tadalafil, and vardenafil are metabolized via CYP3A4, and their effects may be enhanced, or prolonged, in patients who receive inhibitors of this enzyme. Common CYP3A4 inhibitors include macrolide antibiotics, azole antifungals, antiretroviral agents, and grapefruit juice.

CLINICAL CORRELATION

When a patient is taking a PDE-5 inhibitors they should be very cautious of alcohol intake. Alcohol can also cause vasodilation and, when combined with a PDE-5 inhibitor, hypotension can occur putting the patient at risk for fainting/ syncope.

Why should PDE-5 inhibitors and nitrates not be used together?

CASE CONNECTION

BACK TO INTRODUCTION ↑

As you follow up with JD you would make sure to listen carefully to his concerns. The most common medication to treat ED are the PDE-5 inhibitors, however, JD takes a nitrate daily (isosorbide mononitrate). If he takes the nitrate, followed by a PDE-5 inhibitor within 24 hours, it could cause severe hypotension. Recall, PDE-5 inhibitors and nitrate[illegible]reases in cGMP, significant vasodilation, and, ultimately, dramatic reductions in blood pressure. PDE5 inhibitors should therefore not be prescribed to patients receiving any form of nitrate.

Summary

- Nitric oxide (NO) activates guanylyl cyclase, leading to the relaxation of smooth muscles and increased blood flow, facilitating erection.
- Phosphodiesterase type 5 (PDE-5) inhibitors are a common treatment for ED. They work by blocking the enzyme PDE-5, reducing cGMP breakdown, leading to prolonged smooth muscle dilation and increased blood flow in the penis.
- There are four PDE-5 inhibitors available in the United States: sildenafil, tadalafil, vardenafil, and avanafil.
- Common adverse effects of PDE-5 inhibitors include headache, flushing, dizziness, and hypotension due to vasodilation. They can also cause dyspepsia and priapism.
- PDE-5 inhibitors should not be used concurrently with nitrates due to the risk of profound hypotension. They should also be used with caution when combined with other blood pressure lowering agents, alcohol, and CYP3A4 inhibitors.
- Alcohol intake should be monitored when a patient is taking a PDE-5 inhibitor, as it can also cause vasodilation and, when combined with a PDE-5 inhibitor, can lead to hypotension.

Review Questio[illegible]

1. How does dyspepsia occur in association with patient taking PDE-5 inhibitors?

A. ○ Over production of cGMP
B. ○ Over production of nitric oxide
C. ○ Relaxation of corpus cavernosum
D. ● Relaxation of lower esophageal sphincter

Hide Explanation

The correct answer is relaxation of lower esophageal sphincter (D). As PDE-5 inhibitors cause relaxation of smooth muscles it can cause relaxation and gastric acid can reflux into the esophagus, causing dyspepsia. PDE-5 inhibitors do not cause an over production of cGMP (A) nor an overproduction of nitric oxide (B), rather by inhibiting PDE-5 they inhibit the breakdown of cGMP. They do not have an effect on NO. PDE-5 inhibitors do cause relaxation of corpus cavernosum (C), however, this mechanism is related to blood flow in the penis and increasing erectile function making it not related to the adverse effect of dyspepsia.

2. Which PDE-5 inhibitor has the longest duration of action?

A. ○ Sildenafil
B. ● Tadalafil
C. ○ Vardenafil
D. ○ Avanafil

Hide Explanation

The correct answer is Tadalafil (B). Tadalafil has the longest duration of action, up to 36 hours, among the PDE-5 inhibitors. Sildenafil (A), Vardenafil (C), and Avanafil (D) have shorter durations of action.

3. What is the main adverse effect of combining PDE-5 inhibitors with nitrates?

A. ○ Priapism
B. ● Hypotension
C. ○ Flushing
D. ○ Headache
E. ○ Dyspepsia

Hide Explanation

The correct answer is hypotension (B). PDE-5 inhibitors and nitrates act synergistically to cause significant vasodilation and dramatic reductions in blood pressure. Priapism (A) is a potential side effect of PDE-5 inhibitors, but not a result of their combination with nitrates. Flushing (C), headache (D), and dyspepsia (E) are common side effects of PDE-5 inhibitors, but they are not specifically associated with their combination with nitrates.

4. What is the mechanism of action of PDE-5 inhibitors in treating erectile dysfunction?

A. ○ They increase cGMP production
B. ● They block PDE-5 and reduce cGMP breakdown
C. ○ They block nitric oxide
D. ○ They directly cause vasodilation

Hide Explanation

The correct answer is they block PDE-5 and reduce cGMP breakdown (B). PDE-5 inhibitors work by blocking the enzyme PDE-5, which normally breaks down cGMP. This leads to prolonged smooth muscle dilation and increased blood flow in the penis, improving erectile function. They do not directly increase cGMP production (A), nor do they block nitric oxide (C). They do not directly cause vasodilation (D), but rather prolong the vasodilation caused by nitric oxide by preventing the breakdown of cGMP.

(A), nor do they block nitric oxide (C). They do not directly cause vasodilation (D), but rather prolong the vasodilation caused by nitric oxide by preventing the breakdown of cGMP.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action of PDE-5 inhibitors in the context of treating erectile dysfunction.~~
02 ☑ ~~Compare pharmacokinetic properties of PDE-5 inhibitors and their clinical implications.~~
03 ☑ ~~Explain the adverse effects associated with the use of PDE-5 inhibitors.~~
04 ☑ ~~Discuss the drug interactions and contraindications associated with nitrates.~~

Objective Confidence: 4/4